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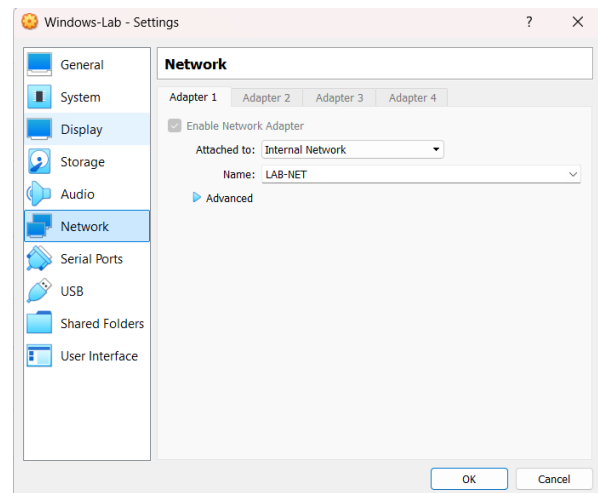
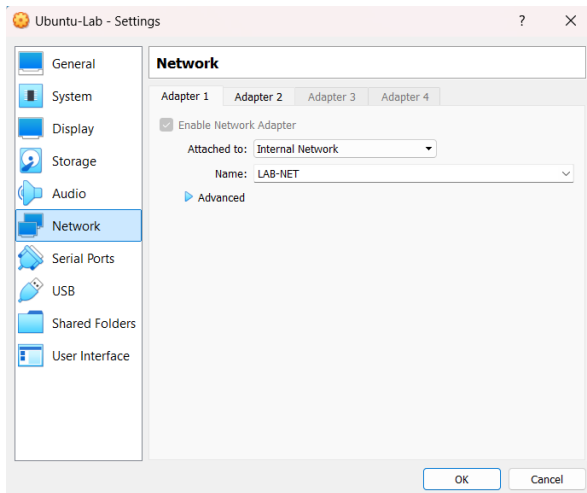
LABORATORY PRACTICE 10

Authentication Log Extraction and Timeline Analysis

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1. Laboratory Network Configuration

The Ubuntu and Windows virtual machines were configured in Oracle VirtualBox with Adapter 1 attached to the Internal Network named LAB-NET. This provided a controlled network path for generating SSH authentication activity between the Windows client and the Ubuntu Linux endpoint.



2. Ubuntu Endpoint Identity Verification

The Ubuntu endpoint identity and network state were verified before generating authentication events. The active account was matthew, the hostname was matthew-VirtualBox, and interface enp0s3 used 10.10.10.10/24. The system time recorded for the activity was September 12, 2026 at 12:37:15 (-05).

```
matthew@matthew-VirtualBox: ~
matthew@matthew-VirtualBox:~$ whoami
matthew
matthew@matthew-VirtualBox:~$ hostname
matthew-VirtualBox
matthew@matthew-VirtualBox:~$ ip -br a
lo                UNKNOWN      127.0.0.1/8 ::1/128
enp0s3            UP          10.10.10.10/24 fe80::6c39:68a2:ee11:51fe/64
enp0s8            UP          10.0.3.15/24 fe80::6119:ea98:ffee:216f/64
matthew@matthew-VirtualBox:~$ date
sáb 12 sep 2026 12:37:15 -05
matthew@matthew-VirtualBox:~$
```

3. SSH Service Verification

The SSH service was checked with systemctl before the remote authentication test. OpenSSH was reported as active and running, with sshd listening on TCP port 22. This confirmed that the Ubuntu endpoint was ready to receive authorized SSH authentication attempts from the Windows laboratory machine.

```
matthew@matthew-VirtualBox:~$ sudo systemctl status ssh
[sudo] contraseña para matthew:
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2026-09-12 12:32:30 -05; 5min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 671 (sshd)
     Tasks: 1 (limit: 4637)
    Memory: 2.5M
   CGroup: /system.slice/ssh.service
           └─671 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups

sep 12 12:32:24 matthew-VirtualBox systemd[1]: Starting OpenBSD Secure Shell server: sshd.
sep 12 12:32:30 matthew-VirtualBox sshd[671]: Server listening on 0.0.0.0 port 22.
sep 12 12:32:30 matthew-VirtualBox sshd[671]: Server listening on :: port 22.
sep 12 12:32:30 matthew-VirtualBox systemd[1]: Started OpenBSD Secure Shell server: sshd.
```

4. Authentication Activity Generation

From Windows PowerShell, an SSH connection was initiated to matthew@10.10.10.10. Two intentionally incorrect password attempts produced Permission denied responses, followed by a successful authentication using the correct password. The successful session opened the Ubuntu shell and confirmed that the authentication activity was generated as intended.

```
Windows PowerShell
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Prueba la nueva tecnología PowerShell multiplataforma https://aka.ms/pscore6

PS C:\Users\matthew> ssh matthew@10.10.10.10
matthew@10.10.10.10's password:
Permission denied, please try again.
matthew@10.10.10.10's password:
Permission denied, please try again.
matthew@10.10.10.10's password:
Welcome to Ubuntu Focal Fossa (development branch) (GNU/Linux 5.4.0-216-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

770 updates can be installed immediately.
488 of these updates are security updates.
Para ver estas actualizaciones adicionales, ejecute: apt list --upgradable

Last login: Tue Sep  8 22:51:49 2026 from 10.10.10.20
matthew@matthew-VirtualBox:~$
```

5. Successful and Failed Event Extraction

The Ubuntu SSH journal was queried with journalctl and filtered with grep for Accepted password and Failed password records. The filtered output showed two failed password events at 12:39:38 and 12:39:45, followed by a successful password event at 12:39:49. Each event identified the matthew account, source address 10.10.10.20, client port 49673, and SSH version 2.

```
matthew@matthew-VirtualBox:~$ sudo journalctl -u ssh --since "10 minutes ago" -o short-iso
| grep -Ei "Accepted password|Failed password"
2026-09-12T12:39:38-0500 matthew-VirtualBox sshd[3876]: Failed password for matthew from 1
0.10.10.20 port 49673 ssh2
2026-09-12T12:39:45-0500 matthew-VirtualBox sshd[3876]: Failed password for matthew from 1
0.10.10.20 port 49673 ssh2
2026-09-12T12:39:49-0500 matthew-VirtualBox sshd[3876]: Accepted password for matthew from
10.10.10.20 port 49673 ssh2
```

6. Recent Authentication Event Review with tail

The same filtered journal output was limited with tail -n 10 to provide a concise view of the most recent matching authentication records. This preserved the successful and failed SSH events in a compact chronological sequence suitable for rapid review during an investigation.

```
matthew@matthew-VirtualBox:~$ sudo journalctl -u ssh --since "10 minutes ago" -o short-iso
| grep -Ei "Accepted password|Failed password" | tail -n 10
2026-09-12T12:39:38-0500 matthew-VirtualBox sshd[3876]: Failed password for matthew from 1
0.10.10.20 port 49673 ssh2
2026-09-12T12:39:45-0500 matthew-VirtualBox sshd[3876]: Failed password for matthew from 1
0.10.10.20 port 49673 ssh2
2026-09-12T12:39:49-0500 matthew-VirtualBox sshd[3876]: Accepted password for matthew from
10.10.10.20 port 49673 ssh2
```

7. Authentication Timeline File

The filtered authentication events were saved to authentication_time.txt and then displayed with cat. Saving the extracted records created a simple evidence artifact that preserved the relevant timestamps and authentication results for later comparison and timeline reconstruction.

```
matthew@matthew-VirtualBox:~$ sudo journalctl -u ssh --since "10 minutes ago" -o short-iso
| grep -Ei "Accepted password|Failed password" | tail -n 10 > authentication_time.txt
matthew@matthew-VirtualBox:~$ cat authentication_time.txt
2026-09-12T12:39:38-0500 matthew-VirtualBox sshd[3876]: Failed password for matthew from 1
0.10.10.20 port 49673 ssh2
2026-09-12T12:39:45-0500 matthew-VirtualBox sshd[3876]: Failed password for matthew from 1
0.10.10.20 port 49673 ssh2
2026-09-12T12:39:49-0500 matthew-VirtualBox sshd[3876]: Accepted password for matthew from
10.10.10.20 port 49673 ssh2
```

8. Authentication Timeline Reconstruction

The extracted records were ordered chronologically to reconstruct the authentication sequence. The timeline contains at least three distinct timestamps and shows the transition from unsuccessful password attempts to a successful SSH login.

Timestamp (-05)	Account	Source IP	Event	Result
2026-09-12 12:39:38	matthew	10.10.10.20	SSH password authentication	Failed
2026-09-12 12:39:45	matthew	10.10.10.20	SSH password authentication	Failed
2026-09-12 12:39:49	matthew	10.10.10.20	SSH password authentication	Successful

9. Observed Authentication Fields and Interpretation

The journal entries provided the key fields required to understand the activity. The same SSH process recorded both the failed and successful attempts, allowing the events to be correlated by account, source address, source port, service, protocol, and time.

Field	Observed Value	Investigation Relevance
Linux endpoint	matthew-VirtualBox / 10.10.10.10	System receiving and recording the SSH authentication activity
Account	matthew	User account involved in the authentication attempts
Source IP	10.10.10.20	Windows laboratory endpoint that initiated the SSH session
Service / process	SSH / sshd	Authentication service that generated the journal records
Destination port	TCP 22	Standard SSH service port
Client source port	49673	Temporary source port recorded in the SSH events
Results	2 Failed password / 1 Accepted password	Shows both unsuccessful and successful authentication outcomes

10. Conclusions

The authentication log analysis was successfully completed on the Ubuntu Linux virtual machine. Controlled SSH activity produced both failed and successful authentication events, which were extracted with journalctl and grep, narrowed with tail, and saved to a timeline file. Three distinct timestamps were identified and ordered to reconstruct the sequence of activity. The exercise demonstrated how Linux authentication logs can be filtered and correlated to identify the user, source address, service, authentication result, and chronological progression of an SSH login sequence.